

A two-stage stochastic multi-objective resilience optimization model for network expansion of interdependent power–water networks under disruption

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ABSTRACT

Network expansion of interdependent critical infrastructures under disruption uncertainty is modeled as a mixed-integer two-stage stochastic multi-objective optimization program. In this model, expected total cost and expected post-disaster resilience are considered competing objectives. Network resilience is quantified through network complexity and unmet demand. Functional relationships between critical infrastructures are modeled using a network-based approach with the physical interdependency enforced through demand constraints. Uncertainty is introduced as a set of random parameters corresponding to disruption scenarios. The proposed model is demonstrated in a case study of coupled power–water networks with the power flow in the grid modeled using linear DC power flow approximation equations. The deterministic-equivalent multi-objective model is solved using the augmented ϵ -constraint method. Solutions from stochastic and deterministic models are compared and the value of stochastic optimization is discussed.

1. Introduction

Many critical infrastructures (CIs) such as power, communications, and transportation systems operate interdependently. They interact and depend on one another to provide essential services to society. The recent example of the Texas power crisis in February 2021 illustrates a clear interdependency between power, gas, and water networks. With the large-scale power failure and interrupted natural gas supplies caused by severe winter storms, the energy crisis transitioned into a water emergency due to low supply and quality issues. The lack of natural gas as a heating source further complicated the extended boil advisory. The impact on the basic needs of 4.5 million residents and the economy was estimated to be at least U.S. \$195 billion [1].

Today's aging CIs not only experience unrelenting demand from economic growth and technological advancement, but their interdependencies also make CIs highly vulnerable to cascading failures. Disruptive events can create widespread losses to critical services that incapacitate the country's physical and economic security. Natural disasters and severe weather dominate over 50% of all events that caused major electrical outages in the US between 2000 and 2016 [2]. The scale of damage to interdependent CIs creates an acute need to strengthen and improve CI resilience to withstand and recover from disruptions.

Optimization models present an approach to influence the decision-making on resilience improvement. Decisions can be differentiated before and after an extreme event as pre-disaster mitigation and post-disaster adaptation, respectively [3]. The pre-disaster approach requires proactive decisions that improve CIs efficiency to withstand loss, e.g., network fortification, asset retrofitting, and network expansion [4, 5]. In contrast, the post-disaster approach requires reactive decisions focusing on the speed of recovery, e.g., task prioritization and resource allocation [6–11]. Strategic planning problems are often associated with proactive decisions while operational and tactical decisions are considered in reactive strategies [12,13]. A combined approach of proactive mitigation and reactive response has also been proposed in a multi-stage decision-making process [14–16].

This research focuses on improving the resilience of interdependent CI systems through the proactive strategy of network expansion which involves adding nodes and links to enhance network redundancy and increase services [17]. The commodity flow allocation can be optimized to meet demands post-disruption using this strategy. We propose that network expansion decisions are influenced by cost as well as resilience measures that are quantified through network performance metrics. The goal is to help network designers make decisions to expand interdependent CI networks through the trade-off analysis between total cost and resilience measures.

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We present a stochastic multi-objective resilience optimization model for network expansion of interdependent CIs under uncertainty. First, we establish a framework to assess a *resilience score* based on the network structural properties and a service-based performance metric post-disruption. With the total cost and the resilience score as two competing objectives, the network expansion problem is formulated using a two-stage approach. The first-stage decisions concern the candidate nodes and links to be established and the second-stage decisions involve flow allocation to meet demands post-disruption. The interdependent relationships between components of CIs and the effect of different disruption scenarios are incorporated. We demonstrate our approach through a real-world example of interdependent power–water networks. The optimal expanded network designs are analyzed with and without uncertainty to illustrate the benefit of the stochastic approach.

This paper contributes to strategic resilience improvement research as follows: (a) we develop a two-stage stochastic multi-objective resilience optimization model for network expansion of interdependent CIs under disruption uncertainty; (b) we propose a resilience score based on network complexity and unmet demand metrics and include it as an additional objective to the total cost in our optimization model; and (c) we demonstrate the model in a real-world case study of power–water networks using the DC power flow approximation [18].

The majority of CIs are privately owned or operated under public governance thereby making joint partnership efforts crucial to resilience improvement strategies [19]. Therefore, the insights and recommendations from this research can potentially benefit cross-sector policymakers when prioritizing investments to strengthen CI resilience.

This paper is organized as follows. In Section 2, existing literature is reviewed and research gaps are identified. Section 3 presents the stochastic multi-objective resilience optimization model formulation for this problem. The interdependent power–water networks case study is presented in Section 4 and computational results are presented in Section 5. Finally, conclusions and future work are provided in Section 6.

2. Literature review

Resilience in the context of CIs is defined as the ability of the systems to absorb the initial shock, adapt to changing conditions, and restore basic functionalities in a timely and efficient manner following a disruption [20–23]. In terms of decision stages, opportunities to improve resilience are distinguished before and after an extreme event [3]. Strategic planning involves pre-disruption decisions made in anticipation of threats, e.g., asset retrofitting and network expansion. In contrast, operational and tactical planning horizons require post-disruption interventions made after the encounter with threats, e.g., recovery efforts, repair crew scheduling, and repair order prioritization. We note that pre-disruption mitigation also benefits the post-disruption adaptation.

CI interdependency is defined as a bidirectional relationship between two CIs in which the state of each infrastructure affects the state of another [19,24]. The interdependencies can be physical, cyber, geographic, logical, or hybrid of each classification [24]. The network perspective is often utilized to present the relationships between components of heterogeneous networks. Namely, the nodes function as generation (supply), consumption (demand), or relay (transshipment) with the links representing the relationships between them [25,26]. Fig. 1 illustrates an example of power and water systems with physical interdependencies, i.e., the output commodity from one CI is required to service another CI through a functional linkage across network boundaries [27]. The interdependent links represent the power demand for equipment operations in the water network and the cooling water supply for power generation.

Network-based models are further classified into topology-based and service-based models [28]. The first approach addresses structural vulnerability through topological characteristics whereas the latter accounts for functional relationships between supply and demand nodes [29,30]. Strategic decisions generally emphasize topology-based metrics since the structural vulnerability dominate the functional counterpart in long-term planning [29]. Additionally, service-based metrics are necessary to model cascading failures that propagate across network elements [26,31,32].

Various resilience assessment frameworks have been proposed based on network performance measures that are either static [33–35], time-dependent [22,25,36,37], or some hybrid approaches [23,38]. Static resilience has been applied in problem domains such as economics, engineering systems, supply chains, and network design [34,35,39–42], whereas dynamic resilience has dominated applications that improve recovery time [34]. Comprehensive reviews of resilience assessment frameworks in the CI context can be found in [22,43], and [23].

Resilience optimization models are developed to evaluate network performance, design and build infrastructure, mitigate damage and loss, and recover system performance [6]. Table 1 summarizes the review of selected literature on resilience optimization models for interdependent CIs. Many applications of interdependent CIs modeling involve power networks due to the high influence on the inoperability of other infrastructures. A large majority of models focus on the reactive strategies of CI recovery and restoration planning post-disruption [7,10,44–47].

Resilience optimization model objectives primarily consist of the total cost related to pre- and post-disaster decisions. Other network performance metrics based on demand satisfaction, network flow, and network efficiency have been used as a resilience measure [5,44,46,47]. A common resilience measure is determined through unmet demand, i.e., the inability to satisfy anticipated demand at demand nodes. For proactive resilience strategies, e.g., network expansion planning [5,38] and defensive resources [48,49], resilience is independent of short-term system dynamics. Due to the complexity of interdependent networks contributing to computational challenges, most studies consider multiple disruption scenarios in a deterministic manner.

A review of the literature shows that the application of multi-objective stochastic program (MSP) for resilience optimization of interdependent CIs is still in a development stage. To the best of our knowledge, our study is the first to consider a two-stage MSP model that improves the resilience of interdependent CIs under disruption uncertainty through network expansion. To model real-world interdependent CIs, network transport equations that represent the physical systems should be incorporated. For power systems, the flow of electricity is governed by the law of physics where the power flow can be manipulated through electrical control devices. The non-linear power flow is commonly approximated as the linear DC power flow to achieve faster convergence in transmission grid expansion planning [18]. We observe that only a small subset of resilience optimization models involving the interdependent power network such as [14,38,50] consider this non-linear effect. Thus, our resilience analysis incorporates the DC power flow approximation to better represent power–water infrastructures.

MSPs can be solved using classical solution methods, e.g., weighted-sum [51], ϵ -constraint method [52,53], modified or augmented ϵ -constraint methods [52,54]. Alternatively, metaheuristics such as evolutionary algorithms and swarm intelligence methods [42,55] have been used to obtain a Pareto front. These solution methods are generally applied to the MSP problem after it is transformed into an equivalent deterministic problem [51].

3. Model formulation

The problem under study is modeled as a single period, two-stage stochastic multi-objective mixed-integer programming (MIP) model. The model analyzes the trade-off between two objectives, i.e., the

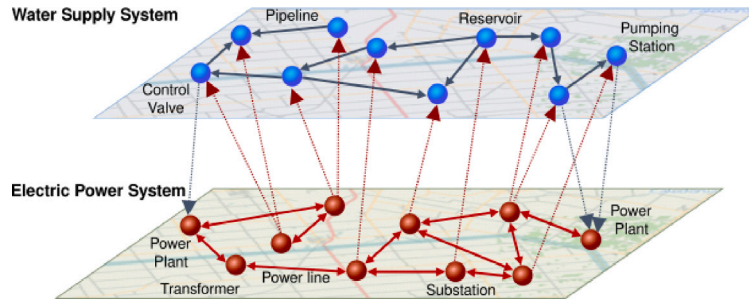


Fig. 1. Example of network-based interdependent electric power–water supply infrastructures.
Source: Adapted from [28].

Table 1
Summary of optimization models for resilience improvement of interdependent infrastructures.

Study	Description	Networks	Resilience strategy	Dynamic vs. static	Resilience measure	Objective(s)	Deterministic vs. stochastic	Solution approach
Lobban et al. [48]	Defense resources allocation planning problem	Power, water, and gas	Proactive	Static	Unmet demand	Minimize vulnerability and minimize total cost	Deterministic	ϵ -constraint method
Byeon et al. [5]	Network expansion planning problem	Power and communication	Proactive	Static	Demand satisfaction	Minimize total cost	Stochastic	Branch-and-price with acceleration schemes
Afrin and Yodo [44]	Hybrid recovery planning based on recovery time and cost-effective strategies	Two generic networks	Reactive	Dynamic	Network performance (efficiency)	Minimize recovery time, then minimize total cost	Deterministic	Exact solution (two stages)
Almoghathawi and Barker [45]	Restoration order prioritization and crew scheduling	Power and water	Reactive	Dynamic	Total met demand over time	Maximize resilience	Deterministic	Exact solution
Karakoc et al. [10]	Restoration order scheduling for community resilience optimization	Power and water	Reactive	Dynamic	Total weighted met demand over time	Maximize community resilience and minimize total cost	Deterministic	ϵ -constraint method
Almoghathawi and Barker [46]	Resource allocation for recovery	Power and water	Reactive	Dynamic	Total flow that reaches demand nodes over time	Maximize resilience	Deterministic	Exact solution
Morshedlou et al. [47]	Binary active and proportional active restoration crew routing problem	Power, water, and gas	Reactive	Dynamic	Total weighted flow that reaches demand nodes over time	Maximize resilience	Deterministic	Relaxed formulation with acceleration schemes
Mooney et al. [7]	Restoration order prioritization, repair crew scheduling, and crew deployment location	Power and water	Reactive	Dynamic	Total met demand over time	Minimize total cost and maximize resilience	Deterministic	ϵ -constraint method
Ouyang [49]	Defender–attacker–defender model for resilience optimization	Power and water	Proactive	Static	Unmet demand	Max–min–max defense/attack/recovery	Deterministic	Decomposition
Ibanez et al. [38]	Network expansion planning problem	Energy and freight transportation	Proactive	Dynamic	Operational cost increases over time	Minimize total cost, minimize CO ₂ emissions, and minimize resilience metrics	Deterministic	NSGA-II evolutionary algorithm
This research	Network expansion planning problem	Power and water	Proactive	Static	Network complexity and unmet demand	Minimize total cost and minimize resilience score	Stochastic	Augmented ϵ -constraint method

total cost of network expansion and the resilience improvement under various disruption scenarios. We measure the resilience improvement through the *resilience score* which accounts for the expanded network complexity and the unmet demand metrics. Solutions identify the set of candidate nodes and candidate links to establish and the redistribution of commodities to meet target performance levels. The proposed model is demonstrated using a real-world case for the integrated power–water infrastructures of Shelby County, TN where the

power flow in transmission line is modeled using the DC power flow approximation.

The functional relationships between components of interdependent CIs are modeled using the network-based approach where nodes correspond to the physical facilities with supply, demand, and transshipment roles. The movement of commodities across network nodes occurs through the directed links that are restricted to one commodity type

Table 2

Sets and parameters for the interdependent network resilience optimization problem.

Sets	
Ω	Set of disruption scenarios ω
K	Set of interdependent infrastructure networks k
R	Set of commodities r in interdependent infrastructure networks
R^k	Set of commodities r in interdependent infrastructure network $k \in K$, $R^k \subseteq R$
$\bar{R}^k$	Set of commodities r not originating from interdependent infrastructure network $k \in K$, $\bar{R}^k \subseteq R^k$
N^k	Set of nodes i in network $k \in K$
N_s^{rk}	Set of supply nodes for commodity $r \in R^k$ in network $k \in K$, $N_s^{rk} \subseteq N^k$
N_t^{rk}	Set of demand nodes for commodity $r \in R^k$ in network $k \in K$, $N_t^{rk} \subseteq N^k$
N_o^{rk}	Set of transshipment nodes for commodity $r \in R^k$ in network $k \in K$, $N_o^{rk} \subseteq N^k$
N_c^k	Set of candidate nodes in network $k \in K$, $N_c^k \subseteq N^k$
A^k	Set of links (i, j) between nodes $i, j \in N^k$; $i \neq j$ in network $k \in K$
$\bar{A}^{k,\omega}$	Set of disrupted links (i, j) between nodes $i, j \in N^k$; $i \neq j$ in network $k \in K$, $\bar{A}^{k,\omega} \subseteq A^k$, for a scenario $\omega \in \Omega$
A_c^k	Set of candidate links (i, j) between nodes $i, j \in N^k$; $i \neq j$ in network $k \in K$, $A_c^k \subseteq A^k$
A^{kl}	Set of interdependent links (i, j) between nodes $i \in N^k$ in network $k \in K$ and $j \in N^l$ in network $l \in K$; $l \neq k$
$\bar{A}^{kl,\omega}$	Set of disrupted interdependent links (i, j) between nodes $i \in N^k$ in network $k \in K$ and $j \in N^l$ in network $l \in K$; $l \neq k$, $\bar{A}^{kl,\omega} \subseteq A^{kl}$, for a scenario $\omega \in \Omega$
A_c^{kl}	Set of candidate interdependent links (i, j) between nodes $i \in N^k$ in network $k \in K$ and $j \in N^l$ in network $l \in K$; $l \neq k$, $A_c^{kl} \subseteq A^{kl}$
Parameters	
p_ω	Occurrence probability of scenario $\omega \in \Omega$
B	Network expansion budget
F_i^k	Fixed cost of node $i \in N_c^k$ in network $k \in K$
f_{ij}^k	Fixed cost of link $(i, j) \in A_c^k$ in network $k \in K$
f_{ij}^{kl}	Fixed cost of interdependent link $(i, j) \in A_c^{kl}$ between networks $k \in K$ and $l \in K$; $l \neq k$
c_{ij}^{rk}	Unit cost of flow on link $(i, j) \in A^k$ for commodity $r \in R^k$ in network $k \in K$
c_{ij}^{rkl}	Unit cost of flow on interdependent link $(i, j) \in A^{kl}$ for commodity $r \in R^k$ in networks $k \in K$ and $l \in K$; $l \neq k$
$s_i^{rk,\omega}$	Supply at node $i \in N^k$ of commodity $r \in R^k$ for network $k \in K$ under scenario $\omega \in \Omega$
$d_i^{rk,\omega}$	Demand at node $i \in N^k$ of commodity $r \in R^k$ for network $k \in K$ under scenario $\omega \in \Omega$
α^r	Minimum percentage of total demand for commodity $r \in R$ to be satisfied
β_i^{rk}	Minimum percentage of demand for commodity $r \in R^k$ to be satisfied at demand node $i \in N_t^{rk}$ in network $k \in K$
$\gamma_i^{rk,\omega}$	Minimum percentage of demand for commodity $r \in \bar{R}^k$ to be satisfied at demand node $i \in N_t^{rk}$ in network $k \in K$
$q_{ij}^{rk,\omega}$	Flow capacity in scenario $\omega \in \Omega$ for link $(i, j) \in A^k$ for commodity $r \in R^k$ in network $k \in K$
$q_{ij}^{rkl,\omega}$	Flow capacity in scenario $\omega \in \Omega$ for interdependent link $(i, j) \in A^{kl}$ for commodity $r \in R^k$ in network $k \in K$ and $l \in K$; $l \neq k$
C_{nco}	Normalization constant for node complexity metric
C_{lco}	Normalization constant for link complexity metric
C_{ud}	Normalization constant for unmet demand metric
w_{co}	Importance weight of network complexity metric in resilience objective function
w_{nco}	Importance weight of node complexity in network complexity metric
w_{lco}	Importance weight of link complexity in network complexity metric
w_{ud}	Importance weight of unmet demand metric in resilience objective function

per link. We assume that the node status is discrete and the non-disrupted demand nodes are only operational if a specific percentage of the anticipated demand is met. The resilience measure is calculated for the expanded network post-disruption. In all disruption scenarios, candidate nodes and links are assumed non-disrupted. Also, the fixed cost of establishing candidate nodes and links, the unit costs of commodity flow in all links, and the supply capacities and demands are assumed to be scenario-independent.

3.1. Stochastic model for the resilient interdependent network expansion problem

Table 2 and Table 3 show the notation used for the model. First-stage decisions are made *here-and-now* before the realization of uncertainty, i.e., the candidate nodes and links to open pre-disaster. Then, based on the disruption scenario realized, the *wait-and-see* decisions in the second stage are conditioned on the network expansion decisions of the first stage. These decisions are the commodity flow allocation, the unused supply, and the unmet demand.

3.2. Objective functions

The model contains two conflicting objectives: expected total cost of network expansion and expected resilience score. Since network expansion decisions are heavily influenced by the total cost, it takes precedence as the first objective function to minimize. Then, resilience score is minimized as the second objective to reduce the network complexity and unmet demand post-disruption.

Disruption scenarios are modeled through uncertain parameters with the uncertainty characterized by a set of finite discrete realizations

of random parameters having a known probability of occurrence. The realizations are represented as a set of scenarios $\{\omega_1, \dots, \omega_n\} \in \Omega$ with corresponding known probabilities $\{p_{\omega_1}, \dots, p_{\omega_n}\}$ and $\sum_{\omega \in \Omega} p_\omega = 1$. The expected values of the objective functions are then optimized over all scenarios.

3.2.1. Expected total cost objective

The expected total cost objective minimizes the overall cost of network expansion, i.e., the fixed cost of adding nodes and links including the expected flow allocation costs over the entire set of disruption scenarios in the second stage. The objective function for the first-stage problem is shown in (1).

$$\begin{aligned} \text{Min}_{\mathbf{y}, \mathbf{z}} Z_1 = & \sum_{k \in K} \left(\sum_{i \in N_c^k} F_i^k y_i^k + \sum_{(i,j) \in A_c^k} f_{ij}^k z_{ij}^k + \sum_{l \in K: l \neq k} \sum_{(i,j) \in A_c^{kl}} f_{ij}^{kl} z_{ij}^{kl} \right) \\ & + \mathbb{E}\{Q(\mathbf{y}, \mathbf{z}, \omega)\} \end{aligned} \quad (1)$$

In (1), $Q(\mathbf{y}, \mathbf{z}, \omega)$ is the optimal value for the objective function of the second-stage problem in (2):

$$Q(\mathbf{y}, \mathbf{z}, \omega) = \text{Min}_{\mathbf{x}, \mathbf{u}, \mathbf{v}} \sum_{k \in K} \sum_{r \in R^k} \left(\sum_{(i,j) \in A^k} c_{ij}^{rk} x_{ij}^{rk,\omega} + \sum_{l \in K: l \neq k} \sum_{(i,j) \in A^{kl}} c_{ij}^{rkl} x_{ij}^{rkl,\omega} \right) \quad (2)$$

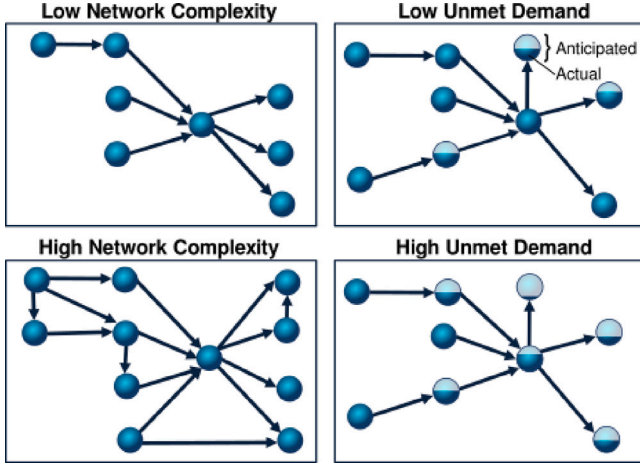
3.2.2. Expected resilience objective

We introduce the *resilience score* that combines topology-based and service-based network performance metrics as a resilience measure. We use the *network complexity* to represent the topology-based metric based on previous supply chain network applications [56–58]. Network complexity consists of the node and link complexities which account

Table 3

Decision variables for the interdependent network resilience optimization problem.

First-stage decision variables	
y_i^k	1 if node $i \in N^k$ in network $k \in K$ is used and 0 otherwise
z_{ij}^k	1 if link $(i, j) \in A^k$ in network $k \in K$ is used and 0 otherwise
z_{ij}^{kl}	1 if interdependent link $(i, j) \in A^{kl}$ in networks $k \in K$ and $l \in K : l \neq k$ is used and 0 otherwise
Second-stage decision variables	
$x_{ij}^{rk,\omega}$	Flow on link $(i, j) \in A^k$ of commodity $r \in R^k$ in network $k \in K$ under scenario $\omega \in \Omega$
$x_{ij}^{rkl,\omega}$	Flow on interdependent link $(i, j) \in A^{kl}$ of commodity $r \in R^k$ in networks $k \in K$ and $l \in K : l \neq k$ under scenario $\omega \in \Omega$
$u_i^{rk,\omega}$	Unused supply at node $i \in N_s^k$ for commodity $r \in R^k$ in network $k \in K$ under scenario $\omega \in \Omega$
$v_i^{rk,\omega}$	Unmet demand at node $i \in N_t^k$ for commodity $r \in R^k$ in network $k \in K$ under scenario $\omega \in \Omega$

**Fig. 2.** Resilience score: network complexity and unmet demand metrics.

for the total number of nodes and links, respectively. The quantity always increases as the network expands. Since a complex network is more vulnerable to high-impact failures, a low network complexity is preferred. For the service-based resilience metric, we use the *unmet demand*, i.e., the difference between the anticipated and the satisfied demands of commodities at demand nodes [42,59]. A network with low unmet demand is preferred since it indicates a high service level. The proposed resilience metrics with low and high levels are illustrated in Fig. 2.

The network complexity and the unmet demand metrics are weighted as shown in the expected resilience objective function presented in (3). For the network complexity metric, only candidate elements are considered in the resilience objective function since the number of existing nodes and links that are non-disrupted will remain fixed. In each scenario, the total unmet demand metric is the sum of unmet demand from both existing non-disrupted demand nodes and the candidate demand nodes that are open.

$$\begin{aligned} \min_{y,z} Z_2 = & w_{co} \left(\sum_{k \in K} \left(\frac{w_{nco}}{C_{nco}} \sum_{i \in N_c^k} y_i^k + \frac{w_{lco}}{C_{lco}} \left(\sum_{(i,j) \in A_c^k} z_{ij}^k + \sum_{l \in K : l \neq k} \sum_{(i,j) \in A_c^{kl}} z_{ij}^{kl} \right) \right) \right) \\ & + \frac{w_{ud}}{C_{ud}} \mathbb{E} \{ \mathcal{W}(y, z, \omega) \} \end{aligned} \quad (3)$$

In (3), $\mathcal{W}(y, z, \omega)$ is the optimal value for the objective function of the second-stage problem (4):

$$\mathcal{W}(y, z, \omega) = \min_{x,u,v} \sum_{k \in K} \sum_{r \in R^k} \sum_{i \in N_t^k} v_i^{rk,\omega} \quad (4)$$

The network complexity metric in (3) is normalized using the total number of nodes and the total number of links in the networks. The unmet demand in (4) is normalized using the total demand of all commodities in the networks. Weight factors w_{co} and w_{ud} are used to prioritize the network complexity and unmet demand metrics, respectively.

3.2.3. Constraints

The constraints associated with the first-stage decisions are defined in (5)–(7). The second-stage constraints are presented in (8)–(20).

$$z_{ij}^k \leq y_i^k \quad \forall (i, j) \in A^k, k \in K \quad (5)$$

$$z_{ij}^{kl} \leq y_i^k \quad \forall (i, j) \in A^{kl}, k \in K, l \in K : l \neq k \quad (6)$$

$$\sum_{k \in K} \left(\sum_{i \in N_s^k} F_i^k y_i^k + \sum_{(i,j) \in A_c^k} f_{ij}^k z_{ij}^k + \sum_{l \in K : l \neq k} \sum_{(i,j) \in A_c^{kl}} f_{ij}^{kl} z_{ij}^{kl} \right) \leq B \quad (7)$$

$$\begin{aligned} & \sum_{(i,j) \in A^k} x_{ij}^{rk,\omega} + \sum_{l \in K : l \neq k} \sum_{(i,j) \in A^{kl}} x_{ij}^{rkl,\omega} - \sum_{(j,i) \in A^k} x_{ji}^{rk,\omega} - \sum_{l \in K : l \neq k} \sum_{(j,i) \in A^{kl}} x_{ji}^{rkl,\omega} \\ & = \begin{cases} s_i^{rk,\omega} - u_i^{rk,\omega} & \forall i \in N_s^k, r \in R^k, k \in K, \omega \in \Omega \\ 0 & \forall i \in N_o^k, r \in R^k, k \in K, \omega \in \Omega \\ -d_i^{rk,\omega} + v_i^{rk,\omega} & \forall i \in N_t^k, r \in R^k, k \in K, \omega \in \Omega \end{cases} \end{aligned} \quad (8)$$

$$x_{ij}^{rk,\omega} \leq q_{ij}^{rk} z_{ij}^k \quad \forall (i, j) \in A^k, r \in R^k, k \in K, \omega \in \Omega \quad (9)$$

$$x_{ij}^{rkl,\omega} \leq q_{ij}^{rkl} z_{ij}^{kl} \quad \forall (i, j) \in A^{kl}, r \in R^k, k \in K, l \in K : l \neq k, \omega \in \Omega \quad (10)$$

$$\sum_{k \in K} \sum_{i \in N_t^{rk}} v_i^{rk,\omega} \leq (1 - \alpha^r) \sum_{k \in K} \sum_{i \in N_t^{rk}} d_i^{rk,\omega} \quad \forall r \in R, \omega \in \Omega \quad (11)$$

$$v_i^{rk,\omega} \leq d_i^{rk,\omega} (1 - y_i^k) + (1 - \beta_i^{rk}) d_i^{rk,\omega} y_i^k \quad \forall i \in N_t^{rk}, r \in R^k \setminus \bar{R}^k, k \in K, \omega \in \Omega \quad (12)$$

$$v_i^{rk,\omega} \leq d_i^{rk,\omega} (1 - y_i^k) + (1 - \gamma_i^{rk}) d_i^{rk,\omega} y_i^k \quad \forall i \in N_t^{rk}, r \in \bar{R}^k, k \in K, \omega \in \Omega \quad (13)$$

$$y_i^k \in \{0, 1\} \quad \forall i \in N^k, k \in K \quad (14)$$

$$z_{ij}^k \in \{0, 1\} \quad \forall (i, j) \in A^k, k \in K \quad (15)$$

$$z_{ij}^{kl} \in \{0, 1\} \quad \forall (i, j) \in A^{kl}, k \in K, l \in K : l \neq k \quad (16)$$

$$x_{ij}^{rk,\omega} \geq 0 \quad \forall (i, j) \in A^k, r \in R^k, k \in K, \omega \in \Omega \quad (17)$$

$$x_{ij}^{rkl,\omega} \geq 0 \quad \forall (i, j) \in A^{kl}, r \in R^k, k \in K, l \in K : l \neq k, \omega \in \Omega \quad (18)$$

$$u_i^{rk,\omega} \geq 0 \quad \forall i \in N_s^k, r \in R^k, k \in K, \omega \in \Omega \quad (19)$$

$$v_i^{rk,\omega} \geq 0 \quad \forall i \in N_t^{rk}, r \in R^k, k \in K, \omega \in \Omega \quad (20)$$

Constraints (5) and (6) are switching constraints relating the status of links to that of the origin nodes. Constraint (7) restricts the expansion to a specific budget. Constraints (8) are the node flow balance constraints for the supply, transshipment, and demand nodes. Constraints (9) and (10) enforce the maximum capacity of the existing and candidate links. Constraints (11) define a specific percentage of the total demand for commodity r that must be met. The relationship between the unmet demand and the status of demand nodes is established in (12) and (13) which enforce the minimum demand of commodity that must be met at the demand node to remain functional. Constraints (13) represent the physical interdependency between networks since the supply nodes of one network act as demand nodes for the interdependent commodity. Finally, constraints (14) to (20) are variable type constraints.

Table 4

DC power flow parameters and variables.

DC power flow parameters	
X_{ij}^E	Series reactance of link $(i, j) \in A^E$ in electrical network E
X_{ij}^{Ek}	Series reactance of interdependent link $(i, j) \in A^{Ek}$ between electrical network E and network $k \in K : k \neq E$
Second stage DC power flow decision variables	
$\theta_i^{E,\omega}$	Voltage phase angle at node $i \in N^E$ in electrical network E under scenario ω
$\mathbf{P}_{ij}^{E,\omega}$	DC power flow on link $(i, j) \in A^E$ in electrical network E under scenario ω
$\mathbf{P}_{ij}^{Ek,\omega}$	DC power flow on interdependent link $(i, j) \in A^{Ek}$ in networks E and $k \in K : k \neq E$ under scenario ω

In summary, the MSP formulation for network expansion of CIs can be represented by its equivalent deterministic formulation as follows:

Minimize $\mathbf{Z}_1$ (1)

Minimize $\mathbf{Z}_2$ (3)

Subject to : (5)–(20)

3.3. DC power flow model

The basic power flow problem requires solving for non-linear voltages, phase angles, and net active and reactive power balances that characterize the sinusoidal behavior of the alternating current (AC) electrical network at steady-state [60]. To model bulk power transfer between substations (i.e., nodes) in the power transmission system, we approximate the non-linear AC system as a linear DC power flow model. The linear approximation is simplified through these assumptions: the power grid only transfers active power, the active power losses in the lines are negligible, the difference in the voltage phase angles are sufficiently small ($\sin(\theta) \approx \theta$ where θ denotes the phase angle measured in radians), and the nominal voltage magnitude is 1.0 per unit [18,60,61].

For simplicity, we define $k = E \in K$ for the electrical network and $r = e \in R$ for electricity. The DC power flow through lines $(i, j) \in A^E$, denoted $\mathbf{P}_{ij}$, can be calculated from (21) [60]. The power flow is a function of the voltage magnitudes at the nodes i and j , the line reactance (X_{ij}), and the difference between the voltage phase angles θ_i and θ_j at nodes i and j , respectively. Transmission lines are bidirectional and a negative $\mathbf{P}_{ij}$ indicates a reverse flow condition from the receiving node j into the sending node i [18,62].

$$\mathbf{P}_{ij} = -\mathbf{P}_{ji} = \frac{(\theta_i - \theta_j)}{X_{ij}} \quad (21)$$

The energy conservation law at node i determines the power leaving the node as $P_i = \sum_{(i,j) \in A^E} \mathbf{P}_{ij} - \sum_{(j,i) \in A^E} \mathbf{P}_{ji}$. The outgoing power $P_i \geq 0$ for power supply nodes, $P_i \leq 0$ for power demand (load) nodes, and $P_i = 0$ for the transshipment nodes [18].

The DC power flow approximation is incorporated as a set of linear constraints using additional parameters and decision variables presented in Table 4. The power flow decision variables are second-stage variables corresponding to the disruption scenario $\omega \in \Omega$.

The DC power flow equation in (21) represents the active power flow in link (i, j) . For all existing and non-disrupted links, the power flow across (i, j) can be modeled through linear constraints (22) and (23).

$$\theta_i^{E,\omega} - \theta_j^{E,\omega} - X_{ij}^E \mathbf{P}_{ij}^{E,\omega} = 0 \quad \forall (i, j) \in A^E \setminus (\bar{A}^{E,\omega} \cup A_c^E), \omega \in \Omega \quad (22)$$

$$\theta_i^{E,\omega} - \theta_j^{E,\omega} - X_{ij}^{Ek} \mathbf{P}_{ij}^{Ek,\omega} = 0 \quad \forall (i, j) \in A^{Ek} \setminus (\bar{A}^{Ek,\omega} \cup A_c^{Ek}), \quad (23)$$

$$k \in K : k \neq E, \omega \in \Omega$$

For candidate and disrupted links, the power flow depends on the link binary decision variables $\mathbf{z}_{ij}^E$ and $\mathbf{z}_{ij}^{Ek}$ as shown in constraints (24) and (25) where M is a very large positive number [63]. If link (i, j) is not established (i.e., $\mathbf{z}_{ij}^E = 0$ or $\mathbf{z}_{ij}^{Ek} = 0$), these constraints allow the voltage angles θ_i and θ_j to take any valid values within the bounds to

satisfy other constraints.

$$|\theta_i^{E,\omega} - \theta_j^{E,\omega} - X_{ij}^E \mathbf{P}_{ij}^{E,\omega}| \leq M(1 - \mathbf{z}_{ij}^E) \quad \forall (i, j) \in (\bar{A}^{E,\omega} \cup A_c^E), \omega \in \Omega \quad (24)$$

$$|\theta_i^{E,\omega} - \theta_j^{E,\omega} - X_{ij}^{Ek} \mathbf{P}_{ij}^{Ek,\omega}| \leq M(1 - \mathbf{z}_{ij}^{Ek}) \quad \forall (i, j) \in (\bar{A}^{Ek,\omega} \cup A_c^{Ek}), \quad (25)$$

$$k \in K : k \neq E, \omega \in \Omega$$

Constraints (24) and (25) are linearized as follows.

$$\theta_i^{E,\omega} - \theta_j^{E,\omega} - X_{ij}^E \mathbf{P}_{ij}^{E,\omega} \leq M(1 - \mathbf{z}_{ij}^E) \quad \forall (i, j) \in (\bar{A}^{E,\omega} \cup A_c^E), \omega \in \Omega \quad (26)$$

$$-\theta_i^{E,\omega} + \theta_j^{E,\omega} + X_{ij}^E \mathbf{P}_{ij}^{E,\omega} \leq M(1 - \mathbf{z}_{ij}^E) \quad \forall (i, j) \in (\bar{A}^{E,\omega} \cup A_c^E), \omega \in \Omega \quad (27)$$

$$\theta_i^{E,\omega} - \theta_j^{E,\omega} - X_{ij}^{Ek} \mathbf{P}_{ij}^{Ek,\omega} \leq M(1 - \mathbf{z}_{ij}^{Ek}) \quad \forall (i, j) \in (\bar{A}^{Ek,\omega} \cup A_c^{Ek}), \quad (28)$$

$$k \in K : k \neq E, \omega \in \Omega$$

$$-\theta_i^{E,\omega} + \theta_j^{E,\omega} + X_{ij}^{Ek} \mathbf{P}_{ij}^{Ek,\omega} \leq M(1 - \mathbf{z}_{ij}^{Ek}) \quad \forall (i, j) \in (\bar{A}^{Ek,\omega} \cup A_c^{Ek}), \quad (29)$$

$$k \in K : k \neq E, \omega \in \Omega$$

The voltage phase angle at node A1 is assigned a reference value of zero as shown in (30). Since the power flow depends on the difference between the voltage phase angles at the two end nodes as shown in (21), all other phase angles will be computed relative to the fixed value at node A1. This reference phase angle is required to make the system of power flow constraints linearly independent and thus solvable [64,65].

$$\theta_{A1}^{E,\omega} = 0, \quad \forall \omega \in \Omega \quad (30)$$

The relative displacement among the voltage phase angles can be either positive or negative depending on the power demand at demand nodes. The bounds of the phase angles are enforced by (31) where π is the mathematical constant representing phase angle in radians [50].

$$-\pi \leq \theta_i^{E,\omega} \leq \pi \quad \forall i \in N^{ek}, k \in K, \omega \in \Omega \quad (31)$$

Power is conserved under steady-state conditions and the total power supply is balanced with the total power demand as shown in (32) [65]. Here, $\mathbf{u}_i^{ek,\omega}$ and $\mathbf{v}_i^{ek,\omega}$ represent the surplus power supply and the unmet demand at node i , respectively.

$$\sum_{i \in N_s^{eE}} (s_i^{eE,\omega} - \mathbf{u}_i^{eE,\omega}) - \sum_{k \in K} \sum_{i \in N_i^{ek}} (d_i^{ek,\omega} - \mathbf{v}_i^{ek,\omega}) = 0 \quad \forall \omega \in \Omega \quad (32)$$

The power supplied, relayed, or demanded at each node is balanced according to constraints (33).

$$\sum_{(i,j) \in A^E} \mathbf{P}_{ij}^{E,\omega} + \sum_{k \in K : k \neq E} \sum_{(i,j) \in A^{Ek}} \mathbf{P}_{ij}^{Ek,\omega} - \sum_{(j,i) \in A^E} \mathbf{P}_{ji}^{E,\omega} - \sum_{k \in K : k \neq E} \sum_{(j,i) \in A^{Ek}} \mathbf{P}_{ji}^{Ek,\omega} = \begin{cases} s_i^{eE,\omega} - \mathbf{u}_i^{eE,\omega} & \forall i \in N_s^{eE}, \omega \in \Omega \\ 0 & \forall i \in N_o^{eE}, \omega \in \Omega \\ -d_i^{ek,\omega} + \mathbf{v}_i^{ek,\omega} & \forall i \in N_i^{ek}, k \in K, \omega \in \Omega \end{cases} \quad (33)$$

Note that the power flows are undirected on link (i, j) and the flow magnitudes $\mathbf{P}_{ij}^k$ and $\mathbf{P}_{ij}^{kl}$ are unrestricted. A positive value indicates that the power flows from the sending node i to the receiving node j whereas a negative value indicates a reverse flow condition. The power

flows $\mathbf{x}_{ij}^{E\omega}$ and $\mathbf{x}_{ij}^{Ek}$: $k \neq E$ used in the variable flow cost in the objective function are the absolute values in (34) and (35).

$$|\mathbf{p}_{ij}^{E\omega}| = \mathbf{x}_{ij}^{E\omega} \quad \forall (i, j) \in A^E, \omega \in \Omega \quad (34)$$

$$|\mathbf{p}_{ij}^{Ek\omega}| = \mathbf{x}_{ij}^{Ek\omega} \quad \forall (i, j) \in A^{Ek}, k \in K : k \neq E, \omega \in \Omega \quad (35)$$

Constraints (34) and (35) are linearized by enforcing constraints (36)–(39).

$$\mathbf{p}_{ij}^{E\omega} \leq \mathbf{x}_{ij}^{E\omega} \quad \forall (i, j) \in A^E, \omega \in \Omega \quad (36)$$

$$\mathbf{p}_{ij}^{E\omega} \geq -\mathbf{x}_{ij}^{E\omega} \quad \forall (i, j) \in A^E, \omega \in \Omega \quad (37)$$

$$\mathbf{p}_{ij}^{Ek\omega} \leq \mathbf{x}_{ij}^{Ek\omega} \quad \forall (i, j) \in A^{Ek}, k \in K : k \neq E, \omega \in \Omega \quad (38)$$

$$\mathbf{p}_{ij}^{Ek\omega} \geq -\mathbf{x}_{ij}^{Ek\omega} \quad \forall (i, j) \in A^{Ek}, k \in K : k \neq E, \omega \in \Omega \quad (39)$$

Variable type constraints for the DC power flow decision variables are shown in (40)–(42).

$$\theta_i^{E,\omega} \text{ unrestricted} \quad \forall i \in N^{ek}, k \in K, \omega \in \Omega \quad (40)$$

$$\mathbf{p}_{ij}^{E,\omega} \text{ unrestricted} \quad \forall (i, j) \in A^E, \omega \in \Omega \quad (41)$$

$$\mathbf{p}_{ij}^{Ek,\omega} \text{ unrestricted} \quad \forall (i, j) \in A^{Ek}, k \in K : k \neq E, \omega \in \Omega \quad (42)$$

The constraints (22)–(23), (26)–(33), and (36)–(42) are added to the general model formulation in Section 3.2.3 to incorporate the DC power flow approximation. The flow conservation constraints in (8) are replaced with (33) in this case.

The stochastic multi-objective MIP formulation is first transformed into an equivalent deterministic multi-objective problem using the approach in [66]. Then, the problem is solved using the augmented ϵ -constraint method [67] to obtain exact solutions using an equivalent deterministic single-objective formulation. The augmented ϵ -constraint method improves from the conventional ϵ -constraint method [68] by generating only the Pareto efficient set and avoiding weakly optimal solutions.

4. Computational experiments: Interdependent power–water networks

The two-stage stochastic multi-objective optimization model for network expansion is demonstrated through a test case of the interdependent power–water networks in Shelby County, TN [69,70]. Expansion of the power network is considered to improve service level under earthquake scenarios.

4.1. Data

Fig. 3 shows the directed network graphs of the interdependent power–water networks of Shelby County, TN. The existing nodes and their connectivities are according to [69,70]. For power network, the out-of-county supplier delivers power at the “Gate Stations” (supply) which relay power through the intersections (transshipment) to the 23 kV and 12 kV substations (demand). The power grid operates on 115–161 kV transmission circuits with a maximum line length of 16 km. Based on the design loads, the reactance per line length of 0.337 Ω/km is used [71,72]. The water network consists of pumping stations and storage tanks as supply nodes. Water demand within each service zone is aggregated into a single node and water is transported by buried pipelines.

Selected power supply nodes receive water from the nearest water nodes for cooling and auxiliary purposes. Similarly, all water pumping stations receive electricity from the nearest power substations for operations and control systems [33]. This accounts for physical interdependencies between the two networks.

Network expansion is planned for the power network only. However, given the interdependency, candidate water links are included to service any candidate power supply nodes that might be opened. The

coordinates of the candidate nodes are randomly generated (Fig. 3(d)). The candidate power node types are randomly assigned and integrated into the existing networks based on the shortest Euclidean distances. We add the origin–destination pairs of candidate links based on the following rules to ensure connectivity: each source node must have at least one outflow, each transshipment node must have at least one outflow and one inflow, and each demand node must have at least one inflow. The supply and demand parameters of candidate nodes are assigned randomly following a uniform distribution to resemble those found in the existing networks. Table 5 summarizes graph properties of the interdependent networks.

Model parameters are shown in Table 6. The fixed cost for each candidate node is assigned according to the node type with supply nodes being the most expensive assets. The fixed cost to open each candidate link is proportional to its length. We consider the expansion budget that is approximately 90% of the full cost of opening all candidate nodes and links. For the expected resilience objective, the component weights w_{co} and w_{ud} for the network complexity and unmet demand metrics are set to emphasize the importance of the service-based network performance. The minimum total service levels of 10% are specified for both infrastructures. At a node level, the demand node requires at least 10% of the anticipated demand to be met to remain operational.

4.2. Disruption scenarios

Shelby County’s infrastructures are susceptible to earthquake hazards [73] and we consider three discrete realizations of earthquake scenarios with magnitudes (M_w) 7.7, 8.0, and 9.0 having an epicenter located at 35°18’N and 90°18’W [6,10,70]. The scenario details are shown in Table 7. While there are common disrupted nodes and links in the three scenarios, the disrupted components in the lower severity scenario are not a complete subset of those in the higher severity scenario. We assume that the scenarios have equal probability. Also, the links that originate from the disrupted nodes will not be functional due to the node being out of service.

5. Computational results and discussion

Computational experiments were performed on an Intel® Xeon® E3-1231 CPU 3.40 GHz with 16.0 GB RAM using Gurobi Optimizer 9.1.2 on Python 3.7.6. The expected total cost is considered the primary objective and the expected resilience is the secondary objective. The MIP problem is solved using augmented ϵ -constraint method with relative gap tolerance ($\leq 1\%$) or time limit ($\geq 28,000$ s) as the termination criteria. Both stochastic and deterministic models are solved to compare the effect of uncertainty on the decisions.

Using the concept of dominance in multi-objective optimization, the Pareto efficient solutions are determined such that the optimal solution is strictly better than other solutions in at least one objective and not worse than other solutions in all objectives. The Pareto solutions are shown in Fig. 4 to illustrate the trade-off between two objectives. For comparison, the expected objective values from the stochastic solutions under three scenarios are plotted with the deterministic ones. All non-dominated solutions in deterministic cases are solved to optimality. However, stochastic solutions fail to converge for the resilience objective values less than 0.6794 due to the time limit exceeded. The relative gap tolerances for the non-converged solutions are between 3.3% to 24.5% for the resilience objective function values between 0.6761 to 0.6558.

In Fig. 4, we see that the resilience score decreases as the total cost increases. Since the lower resilience score indicates a more resilient network design, this validates network expansion as a strategy to enhance post-disruption resilience. For the deterministic solutions, higher total cost are required to maintain the same level of resilience as the disruption impact becomes more severe. This higher cost is a

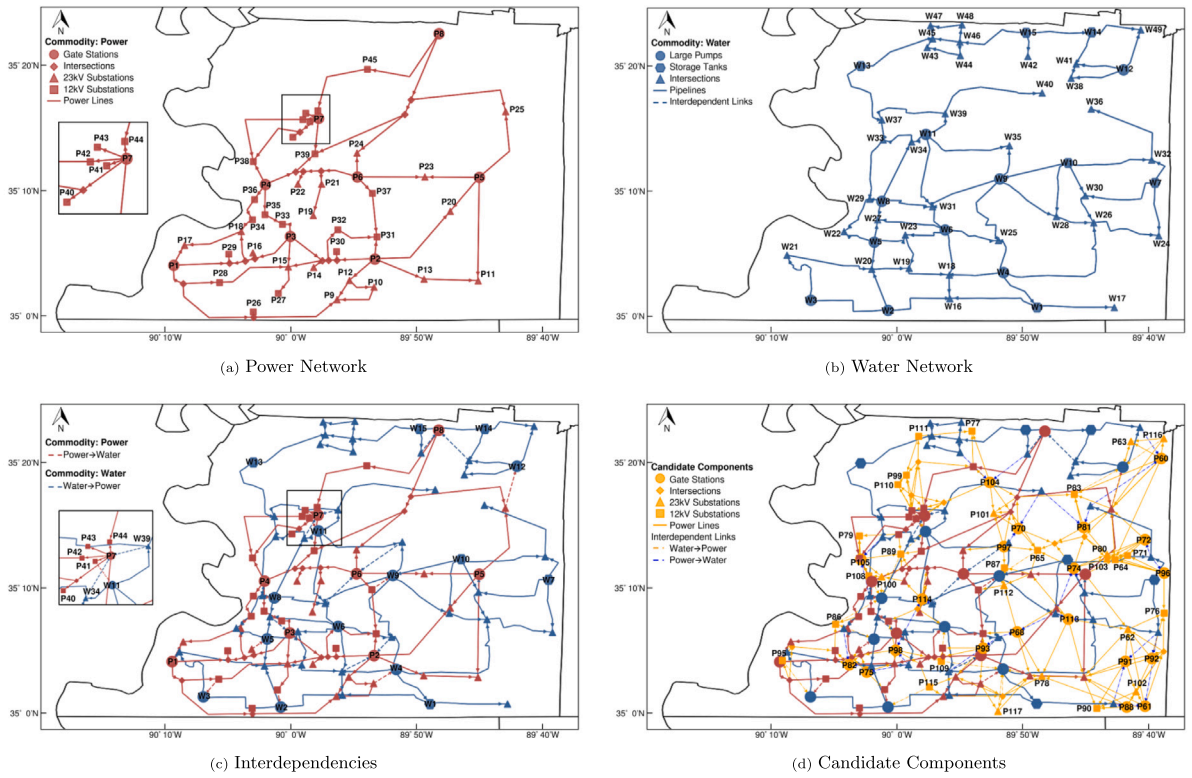


Fig. 3. Interdependent network graphs of the existing (a) power (b) water infrastructures with (c) interdependencies, and (d) candidate components.

Table 5

Graph properties of the interdependent power–water networks.

Network/ Commodity	Components	Node types	Interdependent relationships	No. of nodes		No. of links		Average node degree
				Existing	Candidate	Existing	Candidate	
Power/Electricity	Gate Stations	Power Supply	Demands	8	20	82	222	5.36
	Intersections	Power Transshipment	Utility Water	14	9			
	23 kV Substations	Power Demand		17	11			
	12 kV Substations	Power Demand		20	18			
Water/Utility water	Large Pumps	Water Supply	Demands	9	–	84	35	4.47
	Storage Tanks	Water Supply	Electricity	6	–			
	Intersections	Water Demand		34	–			
Total				166		423		

Table 6

Problem parameters.

Model parameter		Value
w_{co}		0.15
w_{nco}, w_{lco}		0.50
w_{ud}		0.85
$\alpha', \beta_i^k, \gamma_i^k$		10%
Network parameter		Value
Fixed cost	Candidate power node	Supply node: 800, Demand node: $U \sim (200, 600)$, Transshipment node: 400
	Candidate power link	$100+500 \times \text{Length}$; $\text{Length} \in [0.4, 16.0]$
Expansion budget		100,000
Unit flow cost	Power	0.5
	Water	1.0
Supply & Demand	Candidate power supply node	Power supply: $U \sim (70, 290)$, Water demand: 5
	Candidate power demand node	Power demand: $U \sim (4, 80)$
	Existing water supply node	Power demand: 10

Table 7
Disrupted nodes and links in earthquake scenarios.

Scenario	Magnitude (M_w)	No. of disrupted nodes			% of existing nodes	No. of disrupted links			% of existing links	Post-disruption service level	
		Power	Water	Total		Power	Water	Total		Power (Exist.)	Water (Exist.)
1	7.7	7	1	8	7.4%	21	19	40	24.1%	32.8%	79.4%
2	8.0	13	10	23	21.3%	33	29	62	37.3%	26.7%	69.2%
3	9.0	19	15	34	31.5%	43	41	84	50.6%	21.7%	60.9%

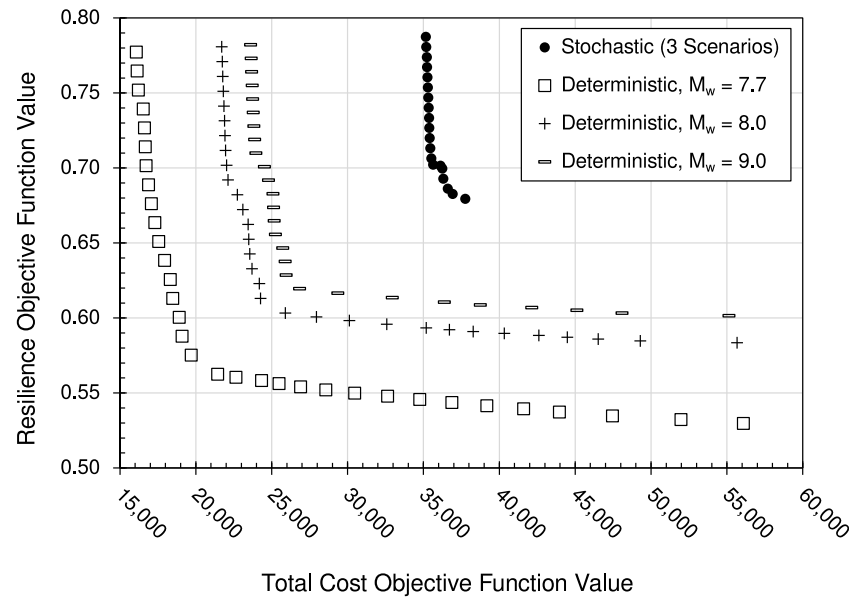


Fig. 4. Deterministic and stochastic Pareto-optimal solutions for power-water network expansion subject to earthquake scenarios.

direct result of adding more nodes and links to supplement commodity supply and capture more demand post-disruption. For stochastic solutions, the expected total cost of maintaining the expected resilience is significantly higher than the deterministic ones and the resilience improvement is more limited. For network expansion problems, the stochastic solutions must accommodate all scenarios considered. Since the disrupted network components in each scenario are not necessary a subset of one another, we see that the stochastic solutions are significantly more pessimistic than the deterministic solutions.

The Pareto solutions from the stochastic model are evaluated using the realization from each disruption scenario and the realized objective functions are plotted in Fig. 5. We observe the expected objective function values as the mathematical expectation of each realization. For each Pareto solution, the resilience objectives have a greater variation than that of the total costs. Thus, the realized resilience score is strongly influenced by the disruption severity.

Since all solutions in the Pareto set are optimal, the decision maker may select a “compromise” solution that satisfies either one of the conflicting objectives. We utilize a multi-criteria decision-making technique called Technique for Order of Preference by Similarity to an Ideal Solution (TOPSIS) [74] to identify such a compromise solution from the Pareto set. The solution with the best relative closeness to the ideal solution, i.e., the shortest distance to the positive-ideal solution and the furthest distance from the negative-ideal solution, is labeled as the TOPSIS solution.

The TOPSIS solutions for both stochastic and deterministic problems are presented in Table 8. Given the higher expected objective values for the stochastic problem, we see that its TOPSIS solution is inferior to the deterministic ones, i.e., a less resilient network comes at a higher cost. Also, the number of candidate nodes and links to open under uncertainty are relatively high given similar resilience criteria. We observe that the fixed cost contributes to the majority of the total cost

in all cases. More supply nodes are added than demand nodes which indicates that the network needs to supply more power to maintain the minimum service level from the higher number of demand nodes post-disruption. Since the transshipment nodes act as a flow-through without capturing additional service, their numbers are limited to at most two in all of the Pareto solutions. Also, the candidate nodes in the stochastic problem are a subset of those found in the deterministic solutions.

Table 9 presents selected solutions from the stochastic Pareto set. Additionally, selected solutions from the three deterministic Pareto sets under earthquake scenarios $M_w = 7.7$, 8.0, and 9.0 are summarized in Table 10. The set of candidate nodes selected in the stochastic solutions (Table 9) are consistent throughout the Pareto set while the optimal candidate locations of the deterministic problem change with different resilience scores (Table 10). This illustrates the first-stage decisions of the stochastic model corresponding to investment decisions that are made “here-and-now” before knowing the disruptions. The power service levels are maintained at a minimum of approximately 10.0% across the Pareto set and the network expands mainly to increase water service. The insight is limited by the availability of converged solutions since the solution method proves to be inefficient at solving for the complete stochastic Pareto set.

In Table 10, we observe increasing total costs with decreasing resilience scores which become prominent as the disruption becomes more severe. The power service level can be improved beyond the minimum (10.0%) in Pareto solutions with the total costs greater than those found in TOPSIS solutions. As a result of more power nodes being added, the network complexity metric becomes more dominant in the resilience score which offsets the reduction in the unmet demand metric. Thus, we see that the return on the fixed cost investment diminishes as the resilience improvement levels off after the TOPSIS total cost. We also found that the candidate nodes associated with the lower costs are not always a subset of those found in the higher-cost

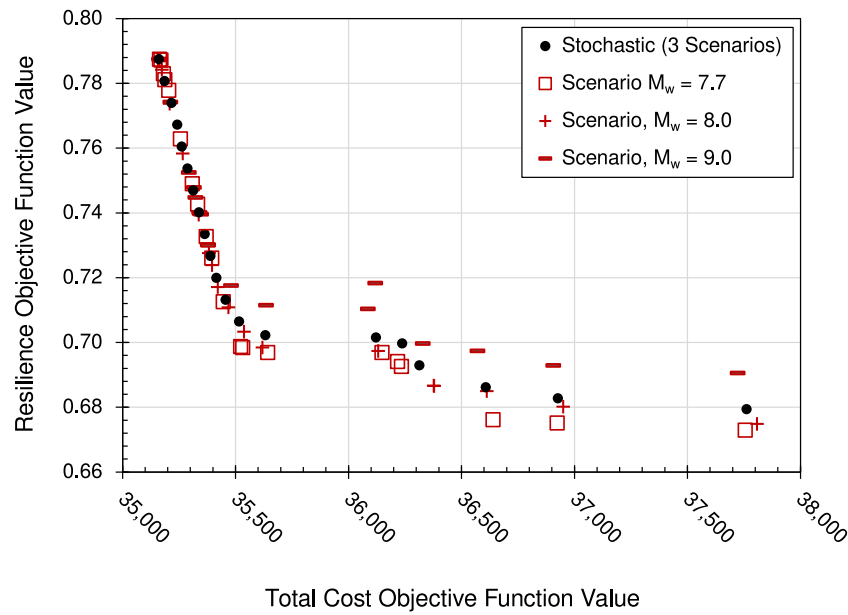


Fig. 5. Expected and realized objective function values from the stochastic Pareto solutions under three disruption scenarios.

Table 8

Summary of TOPSIS solutions under three disruption scenarios.

Scenario	Objectives		Fixed cost	Service level		Candidate nodes open				Candidate links open	
	Total cost	Resilience		Power	Water	Supply nodes		Demand nodes	Trans shipment		Total
Stochastic Problem († Expected value)											
$M_w = 7.7, 8.0, \text{ \& } 9.0$	35,631†	0.7023†	34,958	10.1%†	38.1%†	P60, P61, P68, P70, P75, P82, P88, P91, P92, P93, P97, P98, P113, P114		P62, P63, P65, P76, P78, P87, P90, P101, P102, P116	P69	25	62
Deterministic Problem											
$M_w = 7.7$	18,478	0.6131	17,468	10.0%	64.8%	P82, P93, P97, P98, P105, P113, P114		P79, P87, P95, P100, P108	P69, P94	14	40
$M_w = 8.0$	24,263	0.6131	23,336	10.0%	65.9%	P61, P72, P75, P82, P88, P93, P97, P98, P113, P114		P64, P71, P87, P90, P95, P102, P109	P69	18	49
$M_w = 9.0$	25,953	0.6286	25,070	10.0%	61.4%	P61, P70, P72, P75, P82, P88, P93, P97, P98, P113, P114		P64, P65, P71, P87, P90, P95, P101, P102, P109	P69	21	52

Table 9

Selected Pareto optimal solutions for stochastic model.

Objectives		Fixed cost	Expected network complexity metric	Expected unmet demand Metric	Expected service level		Candidate nodes open					Candidate links open
Expected total cost	Expected resilience				Power	Water	Supply nodes		Demand nodes	Trans shipment	Total	
35,216	0.7740	34,928	0.0223	0.7517	10.1%	14.3%	P60, P61, P68, P70, P75, P82, P88, P91, P92, P93, P97, P98, P113, P114		P62, P63, P65, P76, P78, P87, P90, P101, P102, P116	P69	25	62
35,631 ^a	0.7023	34,958	0.0223	0.6800	10.1%	38.1%	P82, P88, P91, P92, P93, P97, P98, P113, P114		P62, P63, P65, P76, P78, P87, P90, P101, P102, P116	P69	25	62
36,926	0.6827	36,077	0.0225	0.6603	10.1%	44.5%	P97, P98, P113, P114		P101, P102, P116	P69	25	63

^aIndicates TOPSIS solution.

alternatives. The different set of nodes suggests a multi-stage expansion may be beneficial since the capital investments can be spread out over multiple planning periods for the optimal service level over time.

With three disruption scenarios considered in the stochastic problem, the decision maker is expected to invest the total cost of 35,631 for an expected resilience of 0.7023 based on the TOPSIS solution (Table 9). For deterministic cases, the range of total costs to achieve the same resilience score is significantly lower, between 16,713 and 24,588 (Table 10). While having certainty about the disruption enables a much lower investment cost for a comparable resilience score, it is

exceedingly rare for the decision-maker to have complete knowledge of the future. Additionally, obtaining more realistic, meaningful results requires incorporating more disruption scenarios in the stochastic model. Therefore, a more efficient solution method is needed to overcome the computational issues.

5.1. Expanded network and DC power flow results

We select the TOPSIS solution (expected total cost = 35,631, expected resilience 0.7023) to evaluate using the disruption scenario

Table 10

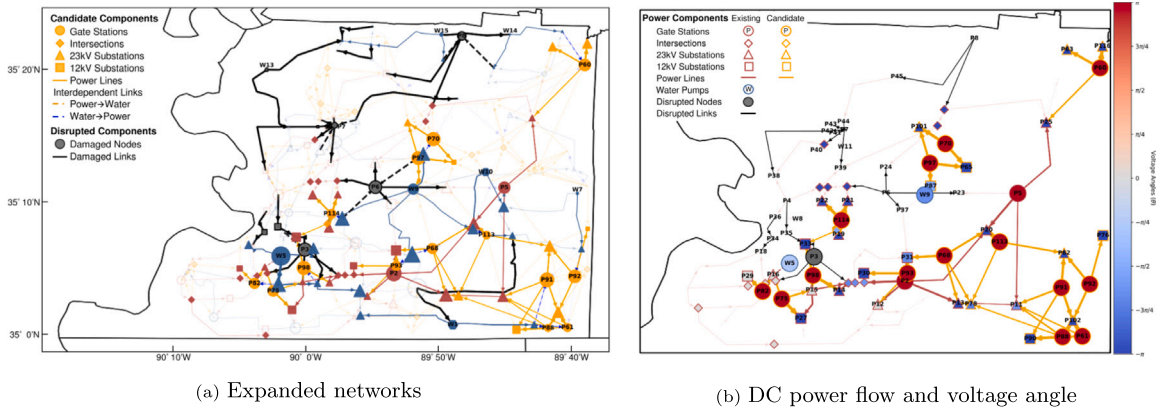
Selected Pareto optimal solution for deterministic model under three earthquake scenarios.

(a) Earthquake scenario $M_w = 7.7$											
Objectives		Fixed cost	Network complexity metric	Unmet demand metric	Service level		Candidate nodes open				Candidate links open
Total cost	Resilience				Power	Water	Supply nodes	Demand nodes	Trans shipment	Total	
16,093	0.7740	15,787	0.0123	0.7617	10.0%	11.1%	P82, P93, P98, P105, P113, P114	P64, P79, P89, P95, P100, P108	P69	13	36
16,713	0.7023	16,156	0.0124	0.6898	10.1%	34.8%	P82, P93, P98, P105, P113, P114	P64, P79, P89, P95, P100, P108	P69	13	37
17,017	0.6827	16,397	0.0126	0.6701	10.0%	41.4%	P82, P93, P97, P98, P105, P113, P114	P79L, P87, P95, P100, P108	P69	13	38
18,478 ^a	0.6131	17,468	0.0134	0.5997	10.0%	64.8%	P82, P93, P97, P98, P105, P113, P114	P79, P87, P95, P100, P108	P69, P94	14	40
19,574	0.5835	18,546	0.0147	0.5688	10.0%	75.0%	P82, P93, P97, P98, P105, P113, P114	P65, P79, P87, P95, P100, P108, P109	P69, P94	16	42
51,965	0.5323	50,674	0.0338	0.4984	18.0%	83.8%	P60, P61, P68, P70, P72, P74, P75, P82, P88, P91, P93, P96, P97, P98, P104, P105, P113, P114	P64, P65, P71, P78, P79, P80, P87, P89, P90, P95, P100, P101, P102, P103, P108, P116	P69, P94	36	99
(b) Earthquake scenario $M_w = 8.0$											
Objectives		Fixed cost	Network complexity metric	Unmet demand metric	Service level		Candidate nodes open				Candidate links open
Total cost	Resilience				Power	Water	Supply nodes	Demand nodes	Trans shipment	Total	
21,727	0.7740	21,466	0.0158	0.7582	10.0%	12.2%	P61, P72, P75, P82, P88, P93, P97, P98, P113, P114	P64, P65, P71, P87, P90, P95, P102	P69	18	43
22,019	0.7023	21,466	0.0158	0.6865	10.0%	36.0%	P61, P72, P75, P82, P88, P93, P97, P98, P113, P114	P64, P65, P71, P87, P90, P95, P102, P116	P69	18	43
22,836	0.6827	22,075	0.0161	0.6666	10.0%	42.5%	P61, P72, P75, P82, P88, P93, P97, P98, P113, P114	P64, P65, P71, P87, P90, P95, P102, P116	P69	18	45
24,263 ^a	0.6131	23,336	0.0168	0.5963	10.0%	65.9%	P61, P72, P75, P82, P88, P93, P97, P98, P113, P114	P64, P71, P87, P90, P95, P102, P109	P69	18	49
55,665	0.5835	54,550	0.0339	0.5496	15.8%	70.8%	P60, P61, P68, P70, P72, P74, P75, P82, P88, P91, P92, P93, P96, P97, P98, P113, P114	P62, P63, P64, P65, P71, P76, P78, P80, P87, P90, P95, P101, P102, P103, P109, P116	P69, P85	35	102
(c) Earthquake scenario $M_w = 9.0$											
Objectives		Fixed cost	Network complexity metric	Unmet demand metric	Service level		Candidate nodes open				Candidate links open
Total cost	Resilience				Power	Water	Supply nodes	Demand nodes	Trans shipment	Total	
23,652	0.7740	23,380	0.0175	0.7564	10.0%	12.8%	P61, P68, P72, P75, P82, P88, P93, P97, P98, P113, P114	P64, P65, P71, P87, P90, P95, P102, P109	P69	20	48

(continued on next page)

Table 10 (continued).

24,588	0.7023	23,904	0.0182	0.6841	10.0%	36.8%	P61, P68, P72, P75, P82, P88, P93, P97, P98, P113, P114	P64, P65, P71, P87, P90, P95, P101, P102, P109	P69	21	49
25,105	0.6827	24,464	0.0184	0.6678	10.0%	43.4%	P61, P70, P72, P75, P82, P88, P93, P97, P98, P113, P114	P64, P65, P71, P87, P90, P95, P101, P102, P109	P69	21	50
25,953 ^a	0.6286	25,070	0.0187	0.6099	10.0%	61.4%	P61, P70, P72, P75, P82, P88, P93, P97, P98, P113, P114	P64, P65, P71, P87, P90, P95, P101, P102, P109	P69	21	52
33,439	0.6131	32,438	0.0218	0.5913	11.4%	65.7%	P60, P61, P68, P72, P75, P82, P88, P91, P92, P93, P97, P98, P113, P114	P62, P65, P71, P87, P90, P95, P102, P109, P116	P69, P85	24	62

^aIndicates TOPSIS solution.Fig. 6. TOPSIS stochastic optimal solution based on scenario $M_w = 7.7$ (realized total cost = 35,641, realized resilience score = 0.6969)

$M_w = 7.7$ and plot the first-stage and second-stage solutions on the interdependent network graph. Fig. 6 shows the expanded network design showing the opened candidate nodes and links (i.e., y_i^k , z_{ij}^k , and z_{ij}^{kl}) and the resulting DC power flow showing the voltage angles and power flow magnitudes ($\theta_{ij}^{k,\omega}$, $P_{ij}^{k,\omega}$, and $P_{ij}^{kl,\omega}$) for scenario $M_w = 7.7$. The total cost is 35,641 and the resilience score is 0.6969 for this scenario. The network expansion decisions include 25 candidate nodes (14 supply, 9 demand, and 1 transshipment) and 62 candidate links open (Fig. 6(a)). The candidate link length distribution has a mean and median of 4.12 km and 3.70 km, respectively, with a maximum length of 11.24 km. Since the candidate link cost is proportional to its length, candidate nodes with shorter candidate links are generally favored. These findings offer an insight into the set of candidate nodes and links that may be used to initialize heuristic solution methods.

Fig. 6(b) shows the resulting voltage phase angle at each power node and the DC power flow along the transmission lines for this solution. The power gate stations contain the highest voltage angle which indicates that they are supply nodes. The voltage angles are the lowest among substations and water pump which acts as demand nodes. The power flow magnitude is directly proportional to the difference between the voltage angles at each power node and inversely proportional to the line reactance (in Ω/km). Since a shorter power line has lower reactance, this results in a higher power flow for the same phase angle difference. This is indicated in Fig. 6(b) using a thicker line width to show the larger power flow magnitude. We note that while shorter power lines may have lower fixed costs, the higher power flow in these lines will contribute to higher flow costs. The DC power flow model is

found to be computationally challenging to solve in comparison to a generic network flow balancing mode.

Fig. 7 presents the optimal network graphs and DC power flows for the deterministic scenarios $M_w = 7.7$, 8.0, and 9.0 based on their TOPSIS solutions. In contrast to the stochastic TOPSIS solution (Fig. 6(a)), different sets of candidate power gate stations and power lines are opened in each deterministic scenario. A set of common supply nodes in both stochastic and deterministic cases are observed, i.e., P82, P93, P97, P98, P113, and P114. As the disruption becomes more severe, the power network is expanded to the east (P72) and southeast corner (P61 and P88) of the County. Unlike the stochastic solution, the power demand in the northeast corner of the county is not captured.

5.2. Value of the stochastic model

Under uncertainty, the expected total cost from the stochastic problem is significantly higher than the deterministic cases for a comparable range of resilience scores. The stochastic model represents the Recourse Problem (RP) where decisions are made *here-and-now*. Alternatively, if we have perfect information about all scenarios in the future, the *wait-and-see* solution is $WS = \mathbb{E}[\min_{\tilde{y}} Z(\tilde{y}, \omega)] = \mathbb{E}[Z(\tilde{y}(\omega), \omega)]$ where $\tilde{y}$ is the vector of decision variables. The maximum “cost” that the decision maker would be expected to pay to have accurate information about the future is the difference between RP and WS solutions. This expected cost of knowing the future with certainty is the *Expected Value of Perfect Information (EVPI)*, i.e., $EVPI = RP - WS$. Table 11 shows the total cost EVPI where the WS solution is the expected value of three scenarios subject to a fixed second objective function value.

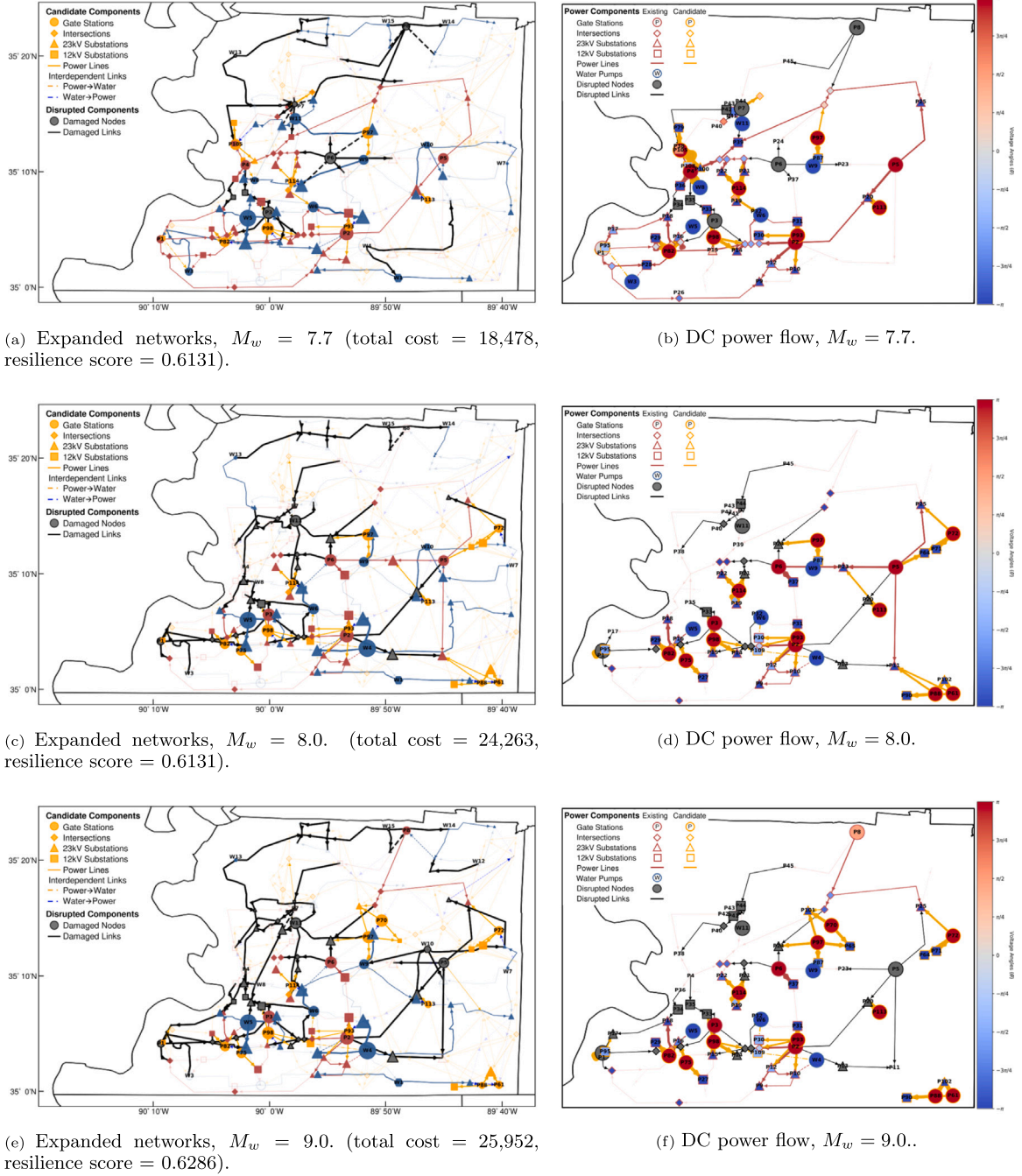


Fig. 7. TOPSIS solution of expanded network design and DC power flow corresponding to three deterministic scenarios.

The *EVPI* in Table 11 shows the performance of the stochastic approach over the deterministic method for the range of resilience scores where the solutions converged. When we make a *here-and-now* decision, the *RP* expected total cost averages 35,799 over the range of resilience scores. If we have perfect information, the average *WS* expected total cost is lower at 20,961. We observe the relation $WS \leq RP$ since the possibility of making the wrong decision is eliminated in absence of uncertainty. The cost of information is significant as the decision maker would need to pay an average *EVPI* of 14,838 instead of knowing the disruption in advance. This corresponds to an $\frac{EVPI}{WS}$ ratio of 70.8% on average. Since the solutions that provide “better” resilience (i.e., lower score) have higher *EVPI*, the decision maker may prefer network designs with lower $\frac{EVPI}{WS}$ ratio given better expected

return on investment. For instance, the lowest ratio of 68.6% can be obtained from solution with $Z_1 = 35,516$ and $Z_2 = 0.7064$ which is relatively close to the TOPSIS solution in this case.

6. Conclusions and future work

This study presents a resilience optimization model for network expansion of interdependent critical infrastructures (CIs). We consider a network-based interdependency between CIs where nodes represent physical assets and links represent functional relationships between them. The operability of individual CI networks is said to be physically interdependent if their operations require commodities from one another.

Table 11
Total cost *EVPI* for various resilience objective function values.

Min Z_2 : Resilience score	Min Z_1 : Total cost			
	RP	WS	EVPI	$\frac{EVPI}{WS}$ (%)
0.7740	35,216	20,490	14,726	71.9%
0.7672	35,241	20,516	14,726	71.8%
0.7605	35,262	20,541	14,720	71.7%
0.7537	35,287	20,572	14,715	71.5%
0.7469	35,312	20,705	14,608	70.6%
0.7402	35,338	20,730	14,608	70.5%
0.7334	35,363	20,756	14,608	70.4%
0.7267	35,389	20,781	14,608	70.3%
0.7199	35,414	20,813	14,602	70.2%

Min Z_2 : Resilience score	Min Z_1 : Total cost			
	RP	WS	EVPI	$\frac{EVPI}{WS}$ (%)
0.7132	35,456	20,848	14,608	70.1%
0.7064	35,516	21,069	14,446	68.6%
0.7023 ^a	35,631	21,107	14,525	68.8%
0.7015	36,121	21,113	15,008	71.1%
0.6997	36,237	21,129	15,108	71.5%
0.6929	36,313	21,203	15,110	71.3%
0.6862	36,607	21,605	15,002	69.4%
0.6827	36,926	21,653	15,273	70.5%
0.6794	37,761	21,671	16,090	74.2%

^aIndicates TOPSIS solution.

We propose a mixed-integer programming formulation of a two-stage stochastic multi-objective optimization model for network expansion under disruption uncertainty. The competing objectives consist of the expected total cost of adding new candidate nodes and links, and the expected resilience score of the expanded network. The resilience score is quantified using network complexity and unmet demand metrics. These topology-based and service-based metrics are used to measure the performance of interdependent networks. The first-stage decisions are the optimal locations of candidate nodes and links to open. The second-stage decisions relate to the optimal flow allocation, the unmet demand, and the unused supply.

Given the influence of power demand by many CI sectors, we present an application of power network expansion and incorporate the DC power flow approximation in the model constraints. Additional second-stage decisions, i.e., the voltage phase angles and the active power flow, are introduced to model the flow mechanism in the power grid. The uncertainty in the stochastic optimization model is introduced as a set of random parameters corresponding to different disruption scenarios.

The model formulation is demonstrated using the case study of the interdependent power–water networks of Shelby County, TN. The physical interdependency is represented by the requirement of electricity for water pump operations and the water supply for cooling purposes at power supply nodes. Three earthquake scenarios with varying degrees of severity are considered with equal probability of occurrence. The stochastic multi-objective MIP model is solved using the augmented ϵ -constraint method to compare the Pareto optimal solutions between stochastic and deterministic problems.

The trade-off between objectives is observed in the increasing total cost to lower the resilience score, i.e., resilience improves with more power nodes and links added to increase service. The Pareto solutions provide insight that post-disruption resilience can be improved significantly with low-cost solutions. Water service level can be increased significantly at a lower cost while power service level requires a much higher cost for a moderate gain. The majority of opened candidate nodes are power supply nodes. Additionally, candidate nodes that require shorter candidate links are favored due to the lower investment cost per length. The DC power flow results show that the higher power flow magnitude in shorter lines contributes to higher flow costs. While we acknowledge the limited number of scenarios tested, it is found that

the expected total cost is significantly greater in the stochastic solution than those obtained from the deterministic solutions given the same resilience level. Solutions for the stochastic model demonstrate that the decision maker is expected to pay a higher total cost without perfect information in the presence of uncertainty.

The augmented ϵ -constraint method used in this study encounters computational challenges in solving the multi-objective stochastic program to optimality within a reasonable time. Considering disruption uncertainty, the stochastic approach would offer a better solution quality if a wider range of scenarios are incorporated. The DC power flow approximation also adds to the complexity of the model formulation and increases the problem size which further compounds the computational challenge. For future work, more efficient solution approaches, e.g., decomposition or metaheuristic methods, should be explored. Since the first-stage decisions are binary and the second-stage decisions are of continuous type, solution methods that exploit the structure of the decision variables should be investigated. The insight from this study can be used to obtain initial guesses of candidate nodes and links in the efficient solution approaches. Based on the observation of a different set of candidate nodes and links across the Pareto set, a multi-stage network expansion planning period can also be considered in future research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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